Summary in Graph

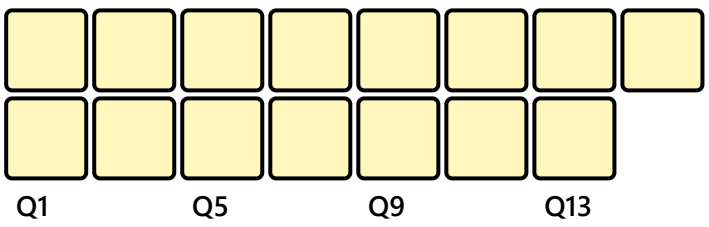
Exam Summary (GO Classes GATE CS | First Order Logic | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	50 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (51) out of 507 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Mathematical Logic

Let $p(x,y), q(x,y)$ denote the following open statements.

$$.\, p(x,y): \quad x^2 \geq y \quad q(x,y): \quad x+2 < y$$

If the universe for each of x, y consists of all real numbers, Which of the following statements is/are false?

A. $p(-3,8) \wedge q(1,3)$
B. $p\left(\frac{1}{2},\frac{1}{3}\right) \vee \neg q(-2,-3)$
C. $p(2,2) \rightarrow q(1,1)$
D. $p(1,2) \leftrightarrow \neg q(1,2)$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Let $\mathbb{Z}$ be the set of integers.
Start with the statement, " $\forall n \in \mathbb{Z}$, if n^2 is even then n is even." Which of the following statements say the same thing?

A. Every integer has an even square and is even.
B. If a given integer is even then that integer has even square.
C. For all integers, some will have an even square.
D. Every integer that has an even square will be even.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Predicates:

- $\mathrm{C(x)}$: x is a CSE 260 student
- $\mathrm{L(x)}$: x loves music

The universe of discourse for the variable x is all students.

Which Translation of the following statements into logical expressions using predicates, quantifiers, and logical connectives is correct?

A. Every student loves music $\forall \mathrm{x L(x)}$
B. No student loves music $\forall \mathrm{x} \neg \mathrm{L(x)}$
C. Some students love music $\exists \mathrm{x L(x)}$
D. Every CSE 260 student loves music. $\forall \mathrm{x(C(x) \rightarrow L(x))}$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Which of the following formulas is a formalization of the sentence:

"Some people are either religious or pious (but not both)"

A. $\exists \mathrm{x(R(x) \oplus P(x))}$
B. $\exists \mathrm{x} \neg \mathrm{(R(x) \leftrightarrow P(x))}$
C. $\neg \forall \mathrm{x[R(x) \leftrightarrow P(x)]}$
D. $\exists \mathrm{x((R(x) \wedge \neg P(x)) \vee (\neg R(x) \wedge P(x)))}$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Which of the following formulas is a formalization of the sentence:

"Not all cars have carburetors."

A. $\neg\forall x[\text{car}(x) \rightarrow \text{carburetors }(x)]$
B. $\forall x[\text{car}(x) \rightarrow \text{carburetors }(x)]$
C. $\exists x[\text{car}(x) \wedge \neg \text{ carburetors }(x)]$
D. $\neg\exists x[\text{car}(x) \wedge \neg \text{ carburetors }(x)]$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

For the following statements, the universe comprises all nonzero integers. Determine the truth value of each statement. Which of the following statements is/are false?

A. $\forall x \exists y[xy = 1]$
B. $\exists x \exists y[(3x - y = 7) \wedge (2x + 4y = 3)]$
C. $\forall y \exists x[xy = 1]$
D. $\exists x \exists y[(2x + y = 5) \wedge (x - 3y = -8)]$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Consider the quantified statement

$$\forall x \exists y[x + y = 17]$$

For which of the following universes, this statement is true?

A. the integers
B. the positive integers
C. the integers for x, the positive integers for y
D. the positive integers for x, the integers for y

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

For which of the following pairs of statements, the proposed negation is the correct negation of the original statement?

A. Statement: For all real numbers x, y, if $x^2 > y^2$, then $x > y$.
Proposed negation: There exist real numbers x, y such that $x^2 > y^2$ but $x \leq y$.
B. Statement: There exist real numbers x, y such that x and y are rational but $x + y$ is irrational.
Proposed negation: For all real numbers x, y, if $x + y$ is rational, then each of x, y is rational.
C. Statement: For all real numbers x, if x is not 0, then x has a multiplicative inverse.
Proposed negation: There exists a nonzero real number that does not have a multiplicative inverse.
D. Statement: There exist odd integers whose product is odd.
Proposed negation: The product of any two odd integers is odd.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let $p(x), q(x)$, and $r(x)$ be the following open statements.

$$\begin{aligned} p(x): &\quad x^2 - 7x + 10 = 0 \\ q(x): &\quad x^2 - 2x - 3 = 0 \\ r(x): &\quad x < 0 \end{aligned}$$

For the universe of all integers, which of the following statements is/are true?

A. $\forall x[p(x) \rightarrow \neg r(x)]$
B. $\forall x[q(x) \rightarrow r(x)]$
C. $\exists x[q(x) \rightarrow r(x)]$
D. $\exists x[p(x) \rightarrow r(x)]$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let $p(x), q(x)$, and $r(x)$ be the following open statements.

$$\begin{aligned} p(x): &\quad x^2 - 7x + 10 = 0 \\ q(x): &\quad x^2 - 2x - 3 = 0 \\ r(x): &\quad x < 0 \end{aligned}$$

For the universe of all positive integers, which of the following statements is/are false?

A. $\forall x[p(x) \rightarrow \neg r(x)]$
B. $\forall x[q(x) \rightarrow r(x)]$
C. $\exists x[q(x) \rightarrow r(x)]$
D. $\exists x[p(x) \rightarrow r(x)]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let $p(x), q(x)$, and $r(x)$ be the following open statements.

$$\begin{aligned} p(x): &\quad x^2 - 7x + 10 = 0 \\ q(x): &\quad x^2 - 2x - 3 = 0 \\ r(x): &\quad x < 0 \end{aligned}$$

If the universe contains only the integers 2 and 5, which of the following statements is/are true?

A. $\forall x[p(x) \rightarrow \neg r(x)]$
B. $\forall x[q(x) \rightarrow r(x)]$
C. $\exists x[q(x) \rightarrow r(x)]$
D. $\exists x[p(x) \rightarrow r(x)]$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

What is the truth of the wff $(\exists x)\,(\mathrm{A}(x) \wedge (\forall y)[\mathrm{B}(x,y) \rightarrow \mathrm{C}(y)])$, where $\mathrm{A}(x)$ is " $x > 0$ " , $\mathrm{B}(x,y)$ is " $x > y$ ", $\mathrm{C}(y)$ is " $y \leq 0$ ", and x is the domain of positive integers and y is the domain of all integers?

A. True
B. False
C. Not a proposition, because there is a free variable in the given wff.
D. Not a proposition, because every variable is a bounded variable in the given wff.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Which of the following is/are true?

A. $[\forall \mathrm{x} \in \mathrm{D}, (\mathrm{P}(\mathrm{x}) \rightarrow \mathrm{Q}(\mathrm{x}))] \rightarrow [(\forall \mathrm{x} \in \mathrm{D}, \mathrm{P}(\mathrm{x})) \rightarrow (\forall \mathrm{x} \in \mathrm{D}, \mathrm{Q}(\mathrm{x}))]$
B. $[\exists \mathrm{x} \in \mathrm{D}, (\mathrm{P}(\mathrm{x}) \rightarrow \mathrm{Q}(\mathrm{x}))] \rightarrow [(\exists \mathrm{x} \in \mathrm{D}, \mathrm{P}(\mathrm{x})) \rightarrow (\exists \mathrm{x} \in \mathrm{D}, \mathrm{Q}(\mathrm{x}))]$
C. $[\forall \mathrm{x} \in \mathrm{D}, (\mathrm{P}(\mathrm{x}) \rightarrow \mathrm{Q}(\mathrm{x}))] \leftarrow [(\forall \mathrm{x} \in \mathrm{D}, \mathrm{P}(\mathrm{x})) \rightarrow (\forall \mathrm{x} \in \mathrm{D}, \mathrm{Q}(\mathrm{x}))]$
D. $[\exists \mathrm{x} \in \mathrm{D}, (\mathrm{P}(\mathrm{x}) \rightarrow \mathrm{Q}(\mathrm{x}))] \leftarrow [(\exists \mathrm{x} \in \mathrm{D}, \mathrm{P}(\mathrm{x})) \rightarrow (\exists \mathrm{x} \in \mathrm{D}, \mathrm{Q}(\mathrm{x}))]$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 11sec | Discuss

Q #14 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Which of the following is/are true?

A. $[\forall \mathrm{x} \in \mathrm{D}, (\mathrm{P}(\mathrm{x}) \Leftrightarrow \mathrm{Q}(\mathrm{x}))] \rightarrow [(\forall \mathrm{x} \in \mathrm{D}, \mathrm{P}(\mathrm{x})) \Leftrightarrow (\forall \mathrm{x} \in \mathrm{D}, \mathrm{Q}(\mathrm{x}))]$
B. $[\exists \mathrm{x} \in \mathrm{D}, (\mathrm{P}(\mathrm{x}) \Leftrightarrow \mathrm{Q}(\mathrm{x}))] \rightarrow [(\exists \mathrm{x} \in \mathrm{D}, \mathrm{P}(\mathrm{x})) \Leftrightarrow (\exists \mathrm{x} \in \mathrm{D}, \mathrm{Q}(\mathrm{x}))]$
C. $[\forall \mathrm{x} \in \mathrm{D}, (\mathrm{P}(\mathrm{x}) \Leftrightarrow \mathrm{Q}(\mathrm{x}))] \leftarrow [(\forall \mathrm{x} \in \mathrm{D}, \mathrm{P}(\mathrm{x})) \Leftrightarrow (\forall \mathrm{x} \in \mathrm{D}, \mathrm{Q}(\mathrm{x}))]$
D. $[\exists \mathrm{x} \in \mathrm{D}, (\mathrm{P}(\mathrm{x}) \Leftrightarrow \mathrm{Q}(\mathrm{x}))] \leftarrow [(\exists \mathrm{x} \in \mathrm{D}, \mathrm{P}(\mathrm{x})) \Leftrightarrow (\exists \mathrm{x} \in \mathrm{D}, \mathrm{Q}(\mathrm{x}))]$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let $\mathrm{Q}(x,y)$ be the statement

"x has been a contestant on television show y."

The domain of discourse for x is the set of all students at your school and for y is the set of all quiz shows on television.
Which of the following are logically equivalent to:

"No student at your school has ever been a contestant on a television quiz show."

A. $\forall x \forall y \cdot \mathrm{NOT}(Q(x,y))$
B. $\exists x \exists y \cdot \mathrm{NOT}(Q(x,y))$
C. $\mathrm{NOT}(\forall x \forall y \cdot Q(x,y))$
D. $\mathrm{NOT}(\exists x \exists y \cdot Q(x,y))$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

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